

FYS4480/9480 Sept 4



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$$N!! = N(N-2)(N-4) \dots 2$$

even

$$N!! = N(N-2)(N-4) \dots 1$$

$$8!! = 8642$$

$$0!! = 1 \quad (-1)!! = 1$$

$$(2n)!! = 2^n n!$$

M = total set of contractions

$$M = 2^n$$

$$x y z \dots n =$$

$$\begin{aligned}
 & N [\overline{x} y z \dots n] + \sum_{(1)} N [\overline{x} y z \dots n] \\
 & \quad \text{contractions} \\
 & + \sum_{(2)} N [\overline{x} y z \dots n] \\
 & \quad + \sum_{(M/2)} N [\overline{x} y z \dots n]
 \end{aligned}$$

Number of operators : 2, 3,
4

Lemma

$$N[x, y, z, \dots, w] \Omega =$$

$$N[x, y, z, \dots, w, e] + \sum_{(1)\Omega} N[x, y, z, \dots, \overbrace{w, e}]$$

$$N[a, q_2] a_e^+$$

$$a_1 q_2 q_e^+ = a_1 (\delta_{e2} - q_e^+ q_2)$$

$$a_1 \delta_{e2} - a_1 q_e^+ q_2 \quad \xrightarrow{-N[a, q_e^+ q_2]}$$

$$\frac{N[a_1 q_2 q_e^+]}{N[a_1 q_2 q_e^+]} - (a_2 \delta_{1e} - q_e^+ a_1 q_2)$$

Wick's Generalized theorem

one-body operator

$$\hat{O}^{(1)} = \sum_{pq} \langle p | \hat{O}^{(1)} | q \rangle \underbrace{a_p^\dagger a_q}_{\text{normal-ordered}}$$

example

$$\hat{H}_0 = \sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q$$

$$\text{if } \hat{h}_0 | q \rangle = \epsilon_q | q \rangle$$

$$\langle p | q \rangle = \delta_{pq}$$

$$= \sum_p \epsilon_p a_p^\dagger a_p$$

two body - operator

$$\hat{O}^{(2)} = \frac{1}{4} \sum_{pqrs} \langle pq | \hat{O}^{(2)} | rs \rangle_{AS} \\ \times a_p^\dagger a_q^\dagger a_s a_r$$

Example

$$\hat{V} = \hat{H}_I = \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \\ \times a_p^\dagger a_q^\dagger a_s a_r$$

Normal-
ordered,

Two-body state

$$|ij\rangle = a_i^\dagger a_j^\dagger |0\rangle$$

$$|kl\rangle = a_k^\dagger a_l^\dagger |0\rangle$$

$$\langle ij | \hat{O}^{(1)} | kl \rangle$$

$$= \sum_{pq} \langle p | \hat{O}^{(1)} | q \rangle \langle 0 | a_j a_i a_p^\dagger a_q^\dagger a_k^\dagger a_l^\dagger | 0 \rangle$$

$$\overbrace{a_i a_j}^+ = 0$$

$$a_i^\dagger a_j^\dagger = 0$$

$$\overbrace{a_i a_j}^- = 0$$

$$\overbrace{a_i a_j^\dagger}^- = \delta_{ij}$$

$$\{a_j a_i\} \{a_p^\dagger a_q^\dagger\} \{a_k^\dagger a_l^\dagger\}$$

Three sets of normal-ordered products of operators

$$a_j \quad a_{i'} \quad a_p^\dagger \quad a_q \quad a_k^\dagger \quad a_l^\dagger$$

$\underbrace{\hspace{1.5cm}} \quad \underbrace{\hspace{1.5cm}} \quad \underbrace{\hspace{1.5cm}}$

$$\delta_{jl} \quad \delta_{i'p} \quad \delta_{qk}$$

$$\underbrace{\delta_{i'p} \delta_{qk}}_{\langle p | \hat{O}^{(i)} | q \rangle \rightarrow \langle i' | \hat{O}^{(i)} | k \rangle}$$

$$\hat{O}^{(i)} | k \rangle = o_k | k \rangle$$

$$\Rightarrow i = k$$

$\hat{O}^{(i)}$ acts on one particle
 at the time, meaning
 that $|i\rangle$ and $|k\rangle$ cannot

differ by more than one
single particle state

$\langle ij | \hat{O}^{(1)} | kl \rangle$ we can
only have non-zero if

$$|kl\rangle = |ij\rangle \quad \text{or}$$

$|kl\rangle$ and $|il\rangle$ differ only
by $k \neq i$

$$|ke\rangle = |i_j\rangle$$

$$\langle i_j | \hat{O}^{(n)} | i_j \rangle$$

$$\sum_{pq} \langle p | \hat{O}^{(n)} | q \rangle \langle 0 | a_j a_i a_p^\dagger a_q a_i^\dagger a_j^\dagger | 0 \rangle$$

$$a_j a_i a_p^\dagger a_q a_i^\dagger a_j^\dagger : \delta_{ip} \delta_{jq} \delta_{ii'} \quad \boxed{\langle i | \hat{O}^{(n)} | i \rangle}$$

$$a_j a_i a_p^\dagger a_q a_i^\dagger a_j^\dagger \quad \delta_{jp} \delta_{iq} \delta_{ii'} \quad \boxed{\langle j | \hat{O}^{(n)} | j \rangle}$$

$$\langle i'j | \hat{O}^{(i)} | ij \rangle = \langle i | \hat{O}^{(i)} | i \rangle + \langle j | \hat{O}^{(i)} | j \rangle$$

if we have $|SD\rangle = |12 \dots N\rangle$

$$= a_1^\dagger a_2^\dagger a_3^\dagger \dots a_N^\dagger |0\rangle$$

$$\langle SD | \hat{O}^{(i)} | SD \rangle = \sum_{k=1}^N \langle k | \hat{O}^{(i)} | k \rangle$$

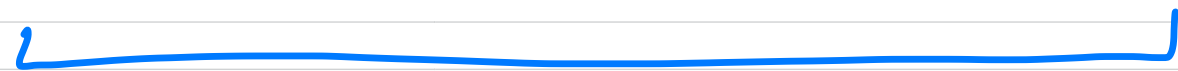
$$\langle ij | \hat{O}^{(1)} | ik \rangle$$

in general we have for an
N-body state

$$\langle SD(j) | \hat{O}^{(1)} | SD(k) \rangle$$

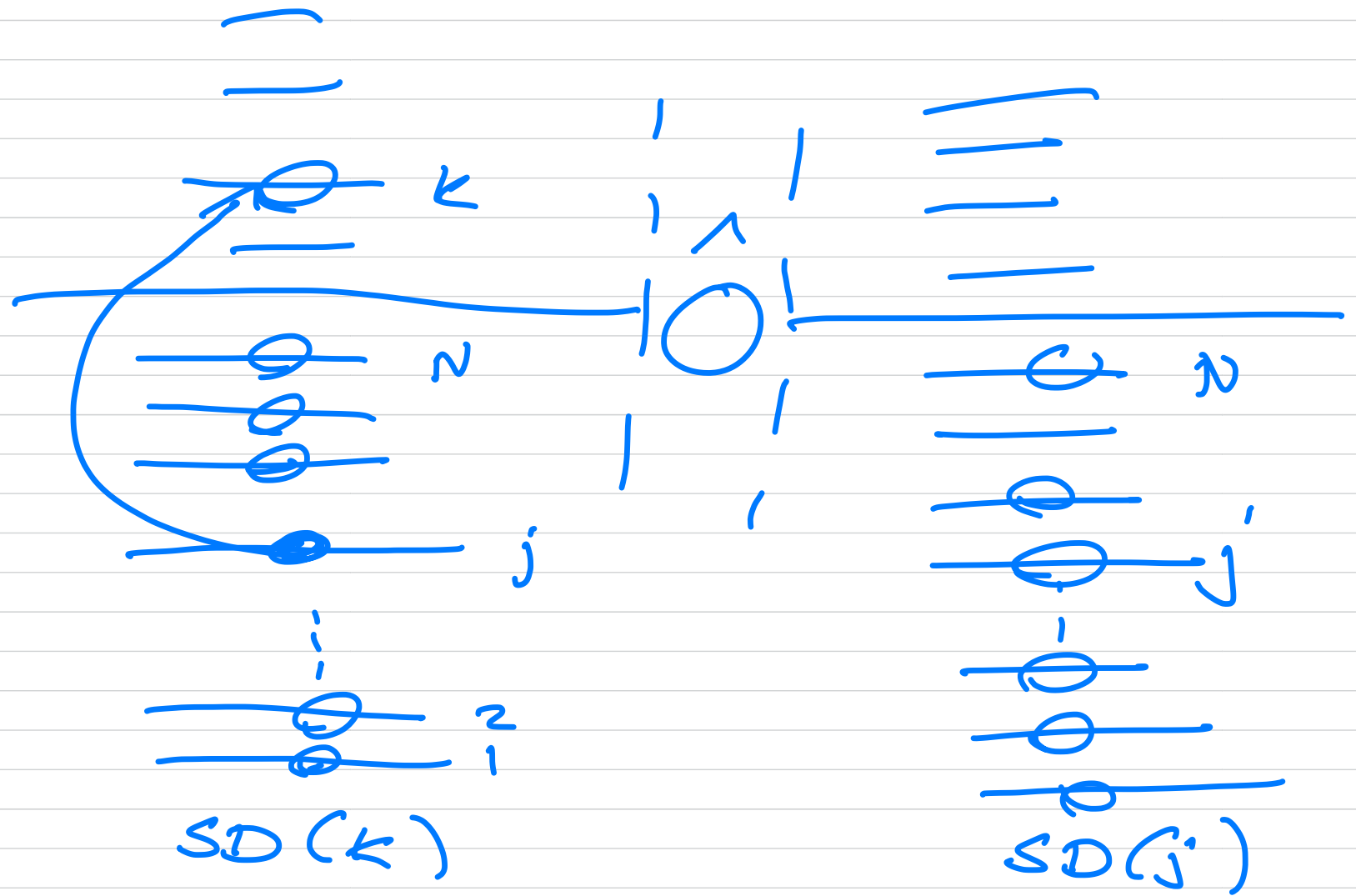
$$\langle SD(j) |$$

$$| 1 2 \dots j \dots N | \hat{O}^{(1)} | 1 2 \dots k \dots N \rangle$$



same slot but

$$j \neq k$$



$$\underbrace{a_j a_{i'} a_p^\dagger a_q}_{\delta_{jk}} \underbrace{a_{i'}^\dagger a_k^\dagger} = 0$$

$$\underbrace{a_j a_{i'} a_p^\dagger a_q}_{\delta_{jk}} \underbrace{a_{i'}^\dagger a_k^\dagger}_{\langle j | k \rangle = 0}$$

$$\delta_{jp} \delta_{qk} \cdot \delta_{i'i} \Rightarrow$$

$$\langle j | \hat{O}^{(1)} | k \rangle$$

$$\langle ij | \hat{O}^{(n)} | ij \rangle = \langle i | \hat{O}^{(n)} | i \rangle + \langle j | \hat{O}^{(n)} | j \rangle$$

same SD $\quad \quad \quad$

$$\langle ij | \hat{O}^{(n)} | ik \rangle = \langle j | \hat{O}^{(n)} | k \rangle$$

SD(j) and SD(k)

Differ by one
sp-state

$$\langle ij | \hat{O}^{(n)} | ke \rangle = 0$$

Differ by two sp-states
SD(ij) SD(ke)

$$a_j, a_1, a_p^\dagger, a_q, a_k^\dagger, a_e^\dagger$$

$$\langle SD | \hat{O}^{(1)} | SD \rangle$$

$$= \sum_{k=1}^N \langle k | \hat{O}^{(1)} | k \rangle$$

$$\langle SD(k) | \hat{O}^{(1)} | SD(j) \rangle$$

$$= \langle k | \hat{O}^{(1)} | j \rangle$$

$$\langle SD(kl) | \hat{O}^{(2)} | SD(ij) \rangle = 0$$

or more than two differ

FCI

$$|\psi_0\rangle = \sum_{j=0}^D c_j \frac{|\Phi_j\rangle}{\sqrt{SD_s}}$$

$\langle \psi_0 | \hat{L}^2 | \psi_0 \rangle$ we will
need elements like

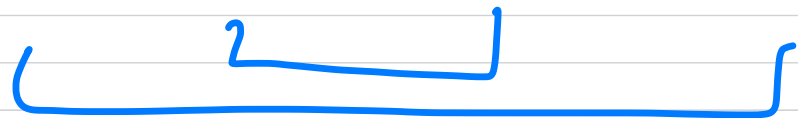
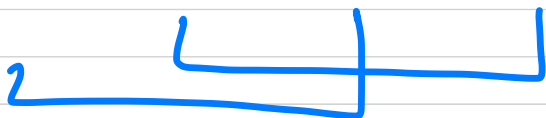
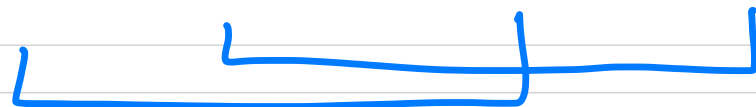
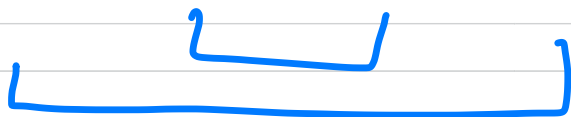
$$\langle \Phi_j | \hat{L}^2 | \Phi_k \rangle$$

$$\hat{O}^{(2)} = \frac{1}{4} \sum_{p \neq r} \langle p q | \hat{O}^{(2)} | r s \rangle_{AS} \\ \times a_p^\dagger a_q^\dagger a_s a_r$$

$$\langle i j | \hat{O}^{(2)} | i j \rangle$$

$$= \frac{1}{4} \sum_{p \neq r} \langle p q | \hat{O}^{(2)} | r s \rangle_{AS}$$

$$\langle c | \{ a_j a_i \} \{ a_p^\dagger a_q^\dagger a_j a_i \} \{ a_i^\dagger a_j^\dagger \} | c \rangle$$

$$q_j \ q_i \ q_p^+ \ q_{\bar{q}}^+ \ q_5 \ q_2 \ q_{\bar{u}}^+ \ q_{\bar{v}}^+$$

 $O^{(2)}$
 $O^{(2)}$

$$\frac{1}{4} \left(\begin{aligned} &\langle ij | \psi | ij \rangle_{A5} - \langle ij' | \psi | ji \rangle_{A5} \\ &+ \langle ji | \psi | ji \rangle_{A5} - \langle ji | \psi | ij \rangle_{A5} \end{aligned} \right) O^{(2)}$$

$$\begin{aligned} \langle i'j | \hat{v} | i'j \rangle &= \langle i'j | \hat{v} | j i' \rangle \\ &= \langle i'j | \hat{v} | i'j \rangle_{AS} \end{aligned}$$

$$\begin{aligned} \langle i'j | \hat{O} | i'k \rangle &= \\ \langle i'j | \hat{O} | i'k \rangle_{AS} \end{aligned}$$

$$\begin{aligned} \langle SD | \hat{O} | SD \rangle &= \\ &= \sum_{i'j} \langle i'j | \hat{O} | i'j \rangle_{AS} \end{aligned}$$

$$\langle SD(j) | \hat{O}^{(2)} | SD(k) \rangle$$

$$= \sum_{i=1}^N \langle i j | \hat{O}^{(2)} | i k \rangle_{AS}$$

$$\underbrace{a_3 a_2 a_1}_{\text{left}} \underbrace{a_p^+ a_q^+}_{\text{middle}} \underbrace{a_5 a_2 a_1}_{\text{right}} \underbrace{a_5^+ a_4^+}_{\text{right}}$$

$$\langle SD(ij) | \hat{O}^{(2)} | SD(kl) \rangle$$

two states that differ

$$= \langle i j | \hat{O}^{(2)} | kl \rangle_{AS}$$

$\langle SD(n|k) | \hat{O}^{(n)} | SD(m|0) \rangle$
 more than two $\neq 0$
 that differ.

$$\begin{array}{ccccccc}
 q_3 & q_2 & q_1 & q_p^+ & q_z^+ & q_v & q_n & q_4^+ & q_5^+ & q_6^+ \\
 & \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & \\
 & & \underbrace{\hspace{2.5cm}} & & & & \underbrace{\hspace{2.5cm}} & & &
 \end{array}$$